



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani Pilani Campus

Academic Undergraduate Studies Division

FIRST SEMESTER 2026-27 COURSE HANDOUT Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course Number	: BIO F214
Course Title	: INTEGRATED BIOLOGY
Instructor-in-Charge	: Pankaj Kumar Sharma [pankajsharma@pilani.bits-pilani.ac.in]
Team of Instructor(s)	: P. Yadukrishnan [yadukrishnan.p@pilani.bits-pilani.ac.in]

1. Course Description: The course intends to bridge the gap and open new vistas to students taking up biology. The course covers two tracks, essentially. The first track introduces the student to the ordering that helps biologists to actually study the vast diversity of the living world. This track would encompass questions related to the origin and evolutionary pathways followed in nature, as well as the methods followed by biologists to systematically categorize and document them. The second track highlights the uses and applications of biology in everyday life – whether in the economic or in the social realms. Together, the course projects the subject in a way from which the student can choose and implement his biological knowledge vis-à-vis his/her interests.

2. Scope and Objective of the Course: Being the second course on General Biology, the course exposes the students to the foundational aspects as described above. At the end of the course, the student will have developed a basic understanding of the evolutionary processes, rationale for taxonomic arrangements and familiarity of selected, representative members of the major kingdoms of living organisms. Further, the student will also become aware of how knowledge of biology (both botany and zoology) is applied for benefit of mankind and other organisms.

3. Course Learning Outcomes:

- a. To understand the scientific basis of evolution in populations
- b. To apply the evolutionary knowledge in classification scheme of organisms

4. Text Books and References:

TB: Raven, P.H. and George B. Johnson. Systematics and Evolutionary Biology (BITS-Pilani Custom Edition 2012). New Delhi: Tata McGraw-Hill Publishing Company Ltd., 2012.

RB1: Campbell, N.A., *et. al.* Essential Biology with Physiology (5th edition). New Delhi: Pearson Education Inc., 2016.

RB2: Starr, Cecie. Biology: Concepts and Applications (6th edition). India: Thomson Brooks/Cole, 2007.



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5. Course Format and Weekly Workflow:

Week No.	Module	Lecture No.* / Description	Reference (Chapter No. from Textbook)	Learning Outcome The student would / would be able to:
1	1. Introduction	L1. Introduction	-	Get introduced to the course
1-2	2. Genes within populations	L2. Genetic variation and evolution, Hardy-Weinberg principle; L3-L4. Agents of evolutionary change; fitness; interaction among evolutionary forces; maintenance of variation; L5. Selection acting on traits; experimental studies on natural selection; limits of selection	20	Identify the forces leading to evolution
3-4	3. Evidence for evolution	L6. Evidence of natural selection; L7. Artificial selection; L8. Fossil and anatomical evidence for evolution; L9. Convergent evolution; Darwin's critics	21	Interpret the pros and cons of Darwin's theory of evolution in light of other theories
4-5	4. Origin of species	L10. The nature of species; the biological species concept; L11-L12. Reproductive isolation; genetic drift and natural selection in speciation; L13-L14. Geography of speciation; adaptive radiation; pace of evolution; speciation and extinction	22	Analyze the processes leading to speciation and extinction
5-6	5. Systematics and the phylogenetic revolution	L15. Systematics; cladistics; L16. Systematics and classification; L17-L18. Phylogenetics and comparative biology; phylogenetics and disease evolution	23	Comprehend and design evolutionary trees
6-7	6. Genome evolution	L19-L20. Comparative genomics; L21. Evolution of whole genomes; L22. Regulatory function of DNA L23. Disease prevention and treatment; crop improvement through genome analysis	24	Compare genomes of organisms at structural and functional levels
8-9	7. Evolution of Development	L24-L25. Overview of evolutionary developmental biology L26-27. New functions for a gene	25	Correlate the evolution of development to life cycle of organisms
10	8. The Tree of Life	L28-L29. Brief overview on classification of organisms (Lecture	26	Classify organisms in



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		cum Self-study)		informative categories based on evolutionary understanding
11	9. Protists	L30. Introduction to protists; origin and endosymbiosis; L31-L32. Protist groups and their economic/medicinal importance	29	Examine similarities and differences among protists in light of evolution
12	10. Green plants and economic botany	L33. Introduction to green algae, bryophytes, tracheophytes; Lycophytes, pteridophytes; L34-L35. Evolution of seed plants; Gymnosperms and angiosperms L36. Application of botanical knowledge for benefit of mankind	30, Class notes	Examine similarities and differences among plants in light of evolution
13	11. Animal Diversity and economic zoology	L37. General features of animals; Evolution of the animal body plan; The classification of animals; specific examples L38. Application of zoological knowledge for benefit of mankind	32, Class notes	Examine similarities and differences among animals in light of evolution
14	12. Epilogue	L39. Concluding the course	-	Connect the different topics learnt in the course to form a coherent summary

* Relevant articles/reading material may be shared from time to time.

6. Evaluation Scheme:

Component	Weightage % (Marks)	Duration	Date & Time	Comments (Closed Book / Open Book)
Mid-Semester Test	30% (60)	90 Min.	05/10 AN1	CB
Comprehensive Examination	40% (80)	3 h	03/12 FN	CB/OB
Quiz(zes) / Assignment(s)*	30% (60)	Variable	-	CB/OB

* A few assignments may be surprise.

Criterion for NC: A student should obtain 30% of the average of the top three performers or 40% of median, whichever is lower, to clear the course.

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Chamber Consultation Hour: Will be announced in the class.



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9. Make-up Policy: Make up for an evaluation component (Mid Sem Test / Compre only) will be considered under extreme and genuine circumstances only such as hospitalization; with prior permission of the IC. Copy of relevant documents needs to be given to the IC to avail makeup. Makeup will not be considered for other components.

10. Others:

Important notices will be provided through email / put up on a shared Google drive folder / Nalanda / announced in class.

For open book exams, only prescribed text book and self-prepared hand-written notes will be allowed.

Ethical standards must be maintained in all evaluation components.

Instructor-in-charge
Course No. BIO F214